1 Question (25 points)

Design a single tape TM that recognizes the language $L = \{0^n 1^{n^2} | n \geq 1\}$. Do not describe the steps of TM. Draw a state diagram, give states, transitions, etc.

2 Question (25 points)

Let $A_n = \{a^k b^m |$ where $k + m$ is a multiple of $n\}$. Show that for each $n \geq 1$, the language is A_n is regular.

3 Question (25 points)

Show that the class of context-free languages is closed under the regular operations,

a. Union

b. Concatenation

c. Star

4 Question (25 points)

Is the following language regular? $B = \{a^n b^m | 0 \leq n < m\}$.